

Soil Moisture Sensor Market Size, Share and Forecast 2036

The global soil moisture sensor market was valued at over USD 566.1 million in 2026 and is anticipated to reach USD 2,350.64 million by the end of 2036, expanding at approximately 15.3% CAGR during the forecast period from 2027 to 2036. Market growth is being driven by increasing precision agriculture, water conservation initiatives, smart irrigation, and the adoption of connected agricultural technologies.

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Soil Moisture Sensor Industry Demand

[Soil moisture sensors](#) are devices designed to measure the amount of water present in soil and provide actionable information for irrigation, crop management, environmental monitoring, and water-resource planning. By delivering real-time or periodic measurements, these sensors enable farmers, researchers, and infrastructure operators to optimize water usage and maintain appropriate soil conditions.

Demand is increasing as agriculture faces water scarcity, unpredictable weather, and the need to improve productivity while minimizing resource consumption. Soil moisture monitoring can reduce unnecessary irrigation, support healthier crop development, and lower operational expenses. Wireless technologies further improve accessibility by enabling remote monitoring and integration with IoT-based farm-management platforms. Long service life, increasingly compact designs, and easier installation are also improving the practicality of sensor-based monitoring systems.

Soil Moisture Sensor Market: Growth Drivers & Key Restraint

Growth Drivers –

- **Expansion of Precision Agriculture:** Farmers are increasingly using data-driven technologies to determine when and where irrigation is required. Soil moisture sensors provide field-level information that supports precise irrigation scheduling and improved crop management.
- **Growing Water Conservation Requirements:** Water scarcity and pressure on agricultural resources are encouraging governments and growers to adopt technologies that minimize water wastage. Sensors can help align irrigation with actual soil conditions rather than fixed schedules.
- **IoT and Sensor Technology Advancements:** Advances in wireless communication, cloud platforms, low-power electronics, and remote sensing are making soil moisture monitoring more scalable and convenient. Integration with automated irrigation systems is further increasing adoption.

Restraint –

- High initial installation costs, sensor calibration requirements, soil-specific measurement variations, and the need for technical expertise can restrict adoption among smaller agricultural operators.

Soil Moisture Sensor Market: Segment Analysis

Segment Analysis by Application

Agriculture: Agriculture represents a major application area because sensors help optimize irrigation, monitor root-zone conditions, and improve resource efficiency. Their integration with precision farming platforms is creating sustained demand.

Environmental Monitoring: Sensors are increasingly deployed to track soil-water conditions, drought patterns, ecosystem changes, and land degradation, supporting environmental research and resource management.

Forestry: Soil moisture monitoring assists forest managers in evaluating water availability, drought stress, wildfire susceptibility, and vegetation conditions across large areas.

Construction & Infrastructure: Sensors can monitor moisture conditions around foundations, embankments, roads, and other structures, helping identify water-related risks and supporting infrastructure management.

Hydrology & Climate Networks: Continuous soil-moisture observations contribute to hydrological models, drought assessment, weather studies, and climate research.

Segment Analysis by Connectivity

Wired Sensors: Wired systems offer stable communication and reliable power connections, making them suitable for permanent installations, research stations, greenhouses, and controlled agricultural environments.

Wireless Sensors: Wireless solutions are experiencing strong adoption because they enable remote monitoring across large or difficult-to-access fields. Connectivity through IoT networks and cloud platforms also supports automated irrigation and real-time decision-making.

Segment Analysis by Sensor Type

Capacitance Sensors: These sensors are widely favored for continuous monitoring because they provide rapid measurements and can be integrated into automated irrigation and precision-agriculture systems.

Time-Domain Reflectometry (TDR): TDR technology provides highly accurate soil-water measurements and is particularly valuable for research, environmental monitoring, and applications requiring reliable data.

Time-Domain Transmission (TDT): TDT sensors provide efficient moisture measurement and are gaining attention where automated and continuous monitoring is required.

Tensiometers: These devices measure soil-water tension and are useful for determining how readily plants can access available moisture, particularly in irrigation management.

Gypsum Blocks: Gypsum-based sensors offer a comparatively simple approach to monitoring soil-water conditions and remain useful in agricultural and research applications where cost considerations are important.

Segment Analysis by End User

Government & Research Institutions: These organizations use soil moisture sensors for agricultural research, hydrology, climate studies, drought monitoring, and environmental programs.

Large-Scale Farms: Commercial farms increasingly adopt sensors to automate irrigation, improve yields, reduce water consumption, and manage large cultivation areas more efficiently.

Irrigation System Integrators: Integrators incorporate sensors into automated irrigation and farm-management systems, creating demand for reliable devices compatible with digital control platforms.

Environmental Agencies: These organizations deploy sensors for watershed monitoring, drought assessment, ecosystem research, and long-term environmental data collection.

Soil Moisture Sensor Market: Regional Insights

North America

North America represents a technologically advanced market supported by precision agriculture, smart irrigation, water-management programs, and extensive adoption of connected farming solutions. Demand is strengthened by commercial farming operations and increasing interest in automated resource management.

Europe

Europe is witnessing increasing demand as sustainable agriculture, water conservation, environmental monitoring, and digital farming become priorities. Government-supported

agricultural modernization and environmental initiatives are encouraging deployment across farms and research networks.

Asia-Pacific (APAC)

Asia-Pacific is emerging as a high-growth market due to expanding agricultural production, water scarcity concerns, modernization of farming practices, and increasing investment in smart agriculture. Adoption is particularly supported by the development of IoT-enabled farming and automated irrigation systems.

Top Players in the Soil Moisture Sensor Market

Key players in the global soil moisture sensor market include METER Group, Inc. (U.S.), Campbell Scientific, Inc. (U.S.), The Toro Company (U.S.), Irrrometer Company, Inc. (U.S.), Delta-T Devices Ltd. (U.K.), Sentek Technologies (Australia), Acclima, Inc. (U.S.), Stevens Water Monitoring Systems, Inc. (U.S.), IMKO Micromodultechnik GmbH (Germany), Spectrum Technologies, Inc. (U.S.), Soilmoisture Equipment Corp. (U.S.), Pessl Instruments GmbH (Austria), Husqvarna Group (Sweden), and CropX Technologies (Israel).

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